

Settings tested: step from rest to 460 RPM, mean of repeats

ID	What changed	Kp	Ki	Filter α	PWM jitter (% duty)	vs baseline	Overshoot (%)	Settling to ± 5 % (s)	RPM std (RPM)	RPM std vs baseline	runs
A	Baseline (original design)	1.56	41.2	1.0 (off)	0.109	—	1.6	0.13	2.17	—	2
C3	Kp 0.8	0.80	41.2	1.0 (off)	0.060	−45 %	1.8	0.07	1.93	−11 %	2
F2	Kp 0.8, Ki 25.0	0.80	25.0	1.0 (off)	0.054	−51 %	1.3	0.19	1.87	−14 %	2
F4	Kp 0.8, Ki 15.0	0.80	15.0	1.0 (off)	0.051	−54 %	1.2	0.36	1.90	−12 %	2
D3	Kp 1.0, α 0.5	1.00	41.2	0.5	0.034	−69 %	10.6	0.12	1.85	−15 %	2
F1	Kp 0.8, α 0.5	0.80	41.2	0.5	0.030	−73 %	13.4	0.15	1.88	−14 %	2
F3	Kp 0.8, Ki 25.0, α 0.5	0.80	25.0	0.5	0.027	−76 %	1.2	0.15	1.87	−14 %	2

PWM jitter = RMS sample-to-sample change of the PWM command over the last 40 % of the run. Blue row = baseline. Row colour = startup overshoot: green ≤ 5 %, amber 5–15 %, red > 15 %.